

International Journal of Science Education

ISSN: 0950-0693 (Print) 1464-5289 (Online) Journal homepage: www.tandfonline.com/journals/tsed20

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To cite this article: Laurence Simonneaux (2001) Role-play or debate to promote students' argumentation and justification on an issue in animal transgenesis, International Journal of Science Education, 23:9, 903-927, DOI: 10.1080/09500690010016076

To link to this article: https://doi.org/10.1080/09500690010016076

Published online: 20 Jul 2010.

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INT. J. SCI. EDUC., 2001, VOL. 23, NO. 9, 903-927

RESEARCH REPORT

Role-play or debate to promote students' argumentation and justification on an issue in animal transgenesis

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This paper compares the impact of a role-play and a conventional discussion on students' argumentation on an issue involving animal transgenesis. Students were confronted with an imaginable but fictional situation. They had to decide whether or not to approve a giant transgenic salmon farm being set up in a seaside village. Students received the same teaching and information, the only differences being in the debate situation. Students were asked to reach a decision on well-argued grounds, to identify areas of uncertainty and to define the condition or conditions under which a change of view might be considered. They had to write them down. Pre-post-tests were used to assess the students' opinions. The role-play and discussion were all video- and audio-taped and transcribed in full. The analysis focuses on the argumentative structure of the students' discourse and identifies the reference areas that students draw on to deliver their arguments. The theory of economics of 'greatness' or 'importance', which has recently emerged as a framework for the sociology of justification, has also been used in analysing students' discourse.

Introduction

Biotechnology is not a scientific discipline as such but a combination of several sciences and techniques. These include cell biology, molecular biology, microbiology, genetics, biochemistry and process engineering as the technological contribution. Biotechnology has different fields of application: agriculture, the chemical, food and pharmaceutical industries, medical diagnostics, the environment and so on. Biotechnology applications raise questions in several areas, including ethics, sociology, economy, ecology, regulation and politics. Debates often lead to moral dilemmas because of uncertainties as to the risks involved. Our aim must be to help students to contribute to the debates, to understand that making decisions can be complex when there are important social issues involved - economic, ethical and so on.

The question of obstacles to the acquisition of knowledge by students in science education needs to be considered on two levels: scientific and technical knowledge on the one hand and opinions on the other. There is little to be said on the former that differs from the statements of science educators about concepts in biology, physics and chemistry, other than to place an even greater emphasis on the incompleteness of knowledge in biotechnology and on the stakes involved in

International Journal of Science Education ISSN 0950-0693 print/ISSN 1464-5289 online © 2001 Taylor & Francis Ltd
http://www.tandf.co.uk/journals
DOI: 10.1080/09500690010016076

research. The second level raises educational questions of a different order. Obstacles to learning in this case are emotional. This does not in any way mean that emotions should be eliminated, but rather that students should be helped to identify their emotional stance as well as the arguments used by scientists, popularizers, teachers, other students and by themselves, together with their validity and the stages involved in reaching a decision.

Our aim must be to help students to identify the criteria and information which support a point of view, theirs as well as those held by others, so that they can treat the issue as problematic. The most effective means of meeting this objective is discussion (in the generic sense). The rules of the game are established and explained, and the objective of the discussion is made clear: this may be to define an issue, to reach a decision on well-argued grounds, to identify areas of uncertainty or to define the condition or conditions under which a change of view may be considered. This sort of situation, in which knowledge is problematic, brings out various facets of problem-solving situations. As well as enacting the different stages in problem finding and problem solving, students placed in a situation in which they have to argue their case are more likely to acquire the knowledge they call on to do so.

By taking part in a debate, students could learn to:

- understand that making decisions can be complex when there are important social issues involved - economic, ethical and so on;
- understand the principles underlying genetic engineering;
- express and defend, or criticize, viewpoints; and
- distinguish between descriptive (describing facts) and normative discourse (evaluating facts) in a discussion.

This kind of situation generates difficulties for students. During arguments, they are potentially in situations of inter or intra subjective conflict. And they are not used to the use of debate in school activities. It also generates difficulties for teachers. They have to change position, to become the manager of debate instead of being 'those who know'.

The aim here was to assess the impact of an educational strategy that was unfamiliar to teachers on the way students come to decisions: the performance of a role-play (variant 1) and comparison with a conventional debate (variant 2) on the same topic (animal transgenesis).

Rationale

This section focuses on the way students make decisions on a given biotechnology application and especially on the arguments they put forward. Argument is a crucial aspect of scientific education (Kuhn 1993). According to Driver and Newton (1997),

> if science is to be taught as socially constructed knowledge then this entails giving a much higher priority than is currently the case to discursive practices in general and to argument in particular. (p. 11)

Several recent studies have addressed the question of students' arguments on the subject of what is sometimes called 'the new genetics' (Waarlo 1998, Zohar and Nemet 1998, Lewis *et al*. 1999).

Some authors hold that argument is an integral part of present-day communication, originating in classic rhetoric (Breton 1996). Others deal with the subject from a linguistic point of view (Adam 1992). Others again address argument at several different levels in science teaching: as a means of improving the understanding of concepts, promoting a better grasp of epistemology in science, developing investigative skills (particularly in practical work), or improving the quality of decision-making on socio-scientific subjects (Geddis 1991, Solomon 1992, Ratcliffe 1996, Driver and Newton 1997, Jiménez Aleixandre *et al*. 1998, Duschl and Ellenbogen 1999, Mortimer and Machado 1999, Osborne 1999).

In analysing different means of persuasion, it is possible to distinguish, in theory at least, between manipulation, propaganda, seduction, argumentation and demonstration. Actually, situations relying purely on seduction or argumentation or demonstration are very rare, a fact which accounts for the development of rhetoric. In France, rhetoric ceased to be taught in schools and universities at the end of the 19th century and even became discredited as a corpus of knowledge. This seems to be linked with a conflict between a culture based on evidence, drawing on advances in science and positivist theory, and a culture based on argumentation (Breton 1996). Interest in rhetoric came to the fore again in the 1960s, with a growing awareness of the impacts of media technology. In 1967, Blanché, in his *Introduction à la logique contemporaine* stated that 'evidence is no longer a guarantee of soundness'.

A new approach in science education discusses the role of rhetoric in teaching and learning (Duschl and Ellenbogen 1999, Mortimer and Machado 1999, Osborne 1999). Despite a vigorous anti-rhetoric tradition, this approach emerged from analyses of observations in classrooms which show that dialogue among students draws on a great deal more than demonstration or explanation - hence a need for rhetoric analysis. These studies on rhetoric in science analyse, *inter alia*, the strategies used in argument and the social procedures that come into play in the validation of scientific knowledge. Osborne (1999) expresses surprise at the apparent lack of interest for rhetoric and argument among science educators, even though these are core aspects of scientific practice. According to Geddis (1991) and Driver *et al*. (1996), this is a significant omission that has had a major impact on science teaching, as it tends to give the wrong impression of the way scientific knowledge is built up. Advances in science are driven by debate and disagreement: there is therefore a need to understand the role of controversy in modern science and its rhetorical functions in handling uncertainty (Osborne 1999). Analyses of argumentation in science education thus also aim to encourage students to develop a critical approach to scientific statements on the repercussions of biotechnology applications.

Studies of argumentation in science education also draw on research in linguistics. Working from a theoretical typology of text sequences, Adam (1992) defines several prototypes: narrative, descriptive, argumentative and explicative sequences, and dialogue.

The following shows how Adam (1992) defines his prototype argument sequence. The pattern of the argument rests on linkages that are made between 'data' and a 'conclusion'. These linkages may be implicitly or explicitly supported (by the 'guarantor' and the 'material') or opposed (by 'refutation' or 'exception').

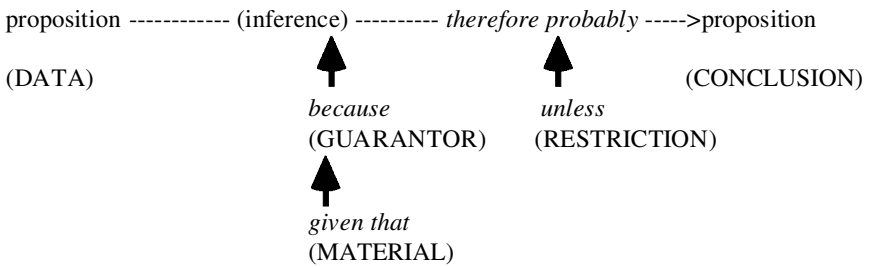


Figure 1. The prototype argument sequence.

Conscientious readers or interlocutors will be attentive to linguistic clues, which lead them to expect some kind of argument. From the point of view of the cognitive treatment of a text, it seems that the understanding of schematic prototype representations that subjects develop over time has an impact on the way they store the information they have considered while coming to grips with a discourse and on the way they look for blocks of information by a process of anticipation (Adam 1992).

Analysing an argumentative text is a new subject in the teaching of French as a first language in upper secondary school. The importance attached to it has to do with its ultimate social purpose, which is related in particular to students' perceptions of citizenship. But this topic is rarely, if ever, taken up in science teaching. Moreover, in French teaching, what is taught is determined by a standard division of descriptive reference theories, and does not touch on the question of values - although this is crucial to the study of argument (Garcia-Debanc, to be published).

Analysing the argumentative structure of students' discursive reasoning and identifying the fields of reference they drew on to produce their arguments does not bring out a full meaningful pattern in data. Students' values are difficult be take into account. This is why we used the economics of 'importance' theory as a framework.

This theory, sometimes also called 'economics of consensus' or 'sociology of justification', has emerged only recently and is still being developed, but it has already made its mark on numerous studies in sociology and economics. It first appeared as a new theoretical and methodological paradigm in a work by Boltanski and Thévenot (1991), entitled '*De la justification. Les économies de la grandeur*'.

The authors consider that there are several 'worlds' in which action takes place and where people call on various skills to cope with their undertakings. This implies that human actions are structured around principles of justification, commitment and criticism. The sociology of justification is a study of discussion and debate. The crucial point in justification processes is the 'orders of importance' that are attached to objects of debate. Agreement between those involved in a debate becomes possible when there is a 'consensus' on these 'orders of importance'.

This theory has aroused a great deal of interest in economics and sociology. A key issue in research that draws on this new theory is discovering how, in an uncertain world, consensus is achieved. The concepts involved in the economics of consensus have also been applied to questions relating to livestock improvement: researchers at the Nationale Institute for Research in Agronomy (INRA) (Vallerand *et al*. 1993), looked into the reasons why Corsican sheep farmers were

refusing to make use of the indices derived from milk yield recording systems and national indexing procedures. They found that the change in the qualification system for breeding animals from a rule-based system involving reputation and custom (domestic consensus) to a system that is guaranteed by genetic procedures (industrial consensus) had given rise to a social crisis. This crisis could only be resolved through 'a compromise over a common good of a higher ''order''. In this particular case, the solution was organised from above around the cultural and heritage value of Corsican sheep (civic consensus)' (Vissac and Vallerand 1993). This type of research highlights the links between a society's acceptance of an innovation (applications of a new form of biotechnology, for instance) and the uncertainty engendered by a change in the consensus underlying relationships between individuals. 'Whether it is accepted or not, managing uncertainty is the key to technology transfer' (Vissac and Vallerand 1993).

But to return to the foundations of the sociology of justification theory. In attempting to achieve a consensus over the perceived 'order of importance' of sociological objects, societies draw on political metaphysics, which are themselves based on political philosophy and political physics. Political philosophy helps to highlight the values (or 'orders of importance') recognised in a society. Political physics sets up rules and laws which are inscribed into what the authors call a 'political grammar' and which define an order of things in each '*city*'. To identify the rules in different 'cities', Boltanski and Thévenot have returned to the 'topical tradition' (from *topoï*: places) and the place of rhetoric (where arguments are found) as defined by Plato, Cicero and Aristotle. They have drawn on the 'canonical texts' which they believe make up the political philosophy of each 'city'. They have thus brought philosophical texts into the test-bed of the sociologist, making it possible to exploit semantic usage and the philosophy of language to clarify the discourse of people in empirical situations (Lafarge 1993).

Boltanski and Thévenot then attempt to define the different 'cities', though without claiming the list to be exhaustive. A 'city' is defined by a set of consensuses and rules that is recognised by each of its members and represents a theoretical model built up from classic works on political philosophy. The concept of a '*common world*' bridges the gap between philosophical and political thinking and pragmatic action.

The 'world of inspiration'. The political philosophy that serves as a reference for the 'city of inspiration' is identified in the work of Saint Augustine. The true greatness of such a city rests on grace (faith), and is opposed to any quest for glory, fame or other form of human dependency. In such a city, imagination, mystery, originality and the invisible are the highest 'orders of importance'.

The 'domestic world'. The underlying philosophy here is drawn from the writings of La Bruyère and Bossuet. Rules are based on reputation and custom. There is an emphasis on the gradual accumulation of qualities through learning and proximity. Fidelity in relationships over time, good will and trust are the foundations of consensus.

The 'world of opinion'. Here, the worth of a person is established through the opinion of others. Boltanski and Thévenot draw in this case on the writings of Hobbes, the 17th century philosopher whose thinking was devoted to the

definition of honour. The truth of an individual's worth cannot be brought into question when it is recognised by others: 'celebrity creates worth'.

The 'world of civics'. To identify the values underlying this 'city', the authors refer to Rousseau's Social Contract. The sovereignty of the 'city' as a whole is disincarnate and depends on the will of all its members. The value of individuals lies in their belonging to a collective being of greater importance. Collective values are instrumentalized and greater than individual interests, with democracy being the most appropriate 'order of importance'.

The 'world of trade'. 'Orders of importance' here are drawn from an analysis of the work of Adam Smith. Competition between economic players, through markets, promotes an optimal use of resources and thus contributes to the overall good of society. Profits are legitimate, and any obstacle to freedom of circulation is prohibited. Quality is assessed through prices.

The 'world of industry'. As developed by Saint-Simon, the political philosophy of the industrial 'city' defines utility as the responses made in order to satisfy needs. The recognized 'order of importance' rests on performance, methods, etc. Material investment has a predominant role. Methods of operation are highly instrumentalized and rules and regulations, standards and technical constraints, have an important place. Standardization of quality is at the core of the 'world of industry'.

Drawing on classic works, Boltanski and Thévenot have defined the attributes of these different 'worlds' as follows: a common higher principle, an 'order of importance', the dignity of individuals, subjects, objects, investment, rewards, images, trials, verdicts and failure. These attributes have different names in each 'world' and are illustrated by words extracted from the corpus of works studied (the words selected by the authors are presented in table 1).

Worlds in conflict

> Everyone has to deal every day with situations that have to do with different worlds, we have to be able to recognise them and to be capable of adjusting to them. Such societies may be described as complex. (Boltanski and Thévenot 1991)

According to the situations in which they find themselves or have put themselves, individuals have to move from one 'world' to another and may therefore recognize several different 'orders of importance'. It is therefore possible to oppose an action or a situation on the grounds represented by an 'order of importance' that has not been previously considered. Disagreement arises when there is no common acknowledgement of an order, or orders, of importance. 'Reality' is therefore not perceived and acknowledged by the different protagonists in the same way. Individuals are thus subject to the constraints of justification, to criticism from different 'worlds' that they have to deal with. To overcome such disagreement, they may reach a compromise between different 'orders of importance'. Compromise rests on mutual arrangements of convenience between protagonists who do not acknowledge a shared 'order of importance'. The need to justify a verdict thus has to be apparent ones actions, and demands tolerance - not as a

Table 1. Author's synopsis of the characteristics of the 'worlds' defined by the economics of importance (Simonneaux 1999)

Worlds	*Inspiration*	*Domestic*	*Opinion*	*Civic*	*Trade*	*Industry*
Common higher principle	Flash of inspiration	Born of tradition	Truth of opinion	Pre-eminence of the collective	Competition	Efficiency
	Inspiration	*Generation, Hierarchy*	*Others, the Public*	*Collective, All, Will*	*Rivalry, Competition*	*Performance Future*
Importance attached to	The inexpressible and ethereal	Superiority of rank	Fame	According to the rules representative	Desirable	Effective
	Out of the ordinary, Emotional	*Benevolence, Faithful, Frank, Trust*	*Renowned, Recognized, Success*	*Unitarian, Legal, Official Representative*	*Value, Millionaire, Winner*	*Functional, Reliable*
Dignity of individuals	Concern for creativeness	The comfort of habit	Desire for recognition	Aspiring to civil rights	Interest	Work
	Love, Passion	*Common sense, What comes naturally, Character*	*Self-esteem, Respect*	*Civil rights, Participation*	*Love (of things), Desire, Egoism*	*Energy*
Subjects	Visionaries	Superiors and inferiors	Stars and their supporters	Organizations and their representatives	Competitors	Professional people
	Spirit, I, Child, Artist	The high: *Father, King, Chief, Family* The low: *Child, Me, Foreigner*	*Personality, Leader, Spokesperson, Media*	*Public bodies, Elected representative, Delegate, Federation, Member*	*Vendor, Customer, Buyer*	*Expert, Manager, Specialist, Agent*

Table 1. ***continued***

Worlds	*Inspiration*	*Domestic*	*Opinion*	*Civic*	*Trade*	*Industry*
Objects	Day-dream	Rules of good behaviour	Names in the media	Legality rights	Wealth	Means to an end
	Spirit, Dream, Unconscious	*Rank, Title, Residence, Gifts*	*Brand name, Press, Decor*	*Legislation, Formality Programme, Policy*	(*luxury*) *objects*	*Tool, Method, Standard, Quantity*
Investment	Escaping from habit	Refusing egoism	Renunication of privacy	Renunciation of the individual	Opportunism	Progress
	Bringing into question. Risk, Detour	*Willingness to help, Duty, Harmony*	*Revealing*	*Solidarity, Overcoming renunciation*	*Freedom, Openings*	*Investment Dynamism*
Rewards	The universal value of the unique	Respect and responsibility	To be recognised and identified	Delegation	Possession	Control
	Genius, Independent	*Authority, Respectability, Honour*	*Identity, Strength*	*Membership Delegation, Representation*		
Images	The reality of imagination	The spirit of home	Public image	Democratic republic	Market	Organization
	Imaginary, Unconscious	*Home, One's environment, Customs*	*Audience, Target*	*Republic, State*		*System*

Trials	Free-ranging spirit	Family ceremonies	Presentation of current events	Demonstration of support to a just cause	Business	Test
	Adventure, Exploration	*Birth, Death, Conversation*	*Event, Conference, Inauguration*	*Assembly, Council, Movement*	*Signed contract*	*Launch, Production*
Verdicts	Stroke of genius	Knowing how trust	The verdict of opinion	The verdict of an election result	Price	Effective
	Illumination, Masterpiece	*Appreciate, Admonition*	*Rumour, Fashion, Popularity*	*Election, Consultation, Mobilization*	*Value (justified, reasonable, right)*	*Correct, Operational*
Failure	Temptation to return to the real world	Rudeness and lack of consideration	Indifference and ordinariness	Division	Slave to money	Instrumental actions
	Habit, External appearance, Reproducing	*Rude, Noisy, Indiscreet, Envious, Perfidious*	*Forgotten, Hidden, Banal, Indifferent*	*In the minority, Isolated, Sectional, Individualism*		*Treating people as objects*

moral value, but as a pragmatic prerequisite in ensuring that things and actions function smoothly.

Students' conceptions on biotechnology may be rooted in different 'worlds', and they may attach different 'orders of importance' to them. Biotechnology is an area that is particularly liable to arouse debate. Whereas in the field of biology education there is little basis for identifying the 'orders of importance' attached to some areas of knowledge being taught (the cell, for example), this is not the case for other topics such as evolution, sexuality or innate versus acquired characteristics. However, it is always central where biotechnology teaching is concerned. 'Orders of importance' indicate the foundations of people's conceptions.

Devising the role-play

As members of the European Initiative for Biotechnology Education (EIBE), we are working on ways of developing the content of biotechnology education for the purpose of teaching today's students, as future actors in the 21st century. EIBE is a group of experts in biotechnology education from 17 European countries working under the aegis of the European Commission. EIBE has several aims, including the development of continuing education in biotechnology among teachers of scientific and non scientific subjects; exploring cultural differences (practices, beliefs and values) among teachers and students and analysing the ways in which these factors can influence perception, acceptance and development of biotechnology at national and European levels; and fostering debate in society by developing forms of education that incorporate the personal, social, ethical, economic and environmental implications of biotechnology.

This has been contributing to the development of teaching material on animal transgenesis (Simonneaux 1997a). The document comprises three modules. The first focuses on the production of transgenic ewes secreting a human enzyme in their milk for therapeutic purposes (alpha 1-proteinase inhibitor) - the ewe called Tracy symbolizes the concept of producing transgenic animals as live fermenters of recombinant proteins. The second module deals with the production of transgenic mice expressing oncogenes to be used as models for cancer studies. The third focuses on transgenic animals in livestock breeding. Most of the goals of animal transgenesis are medical or zootechnic. In view of our mandate, the design of the latter module was entrusted to Ecole Nationale de Formation Agronomique (ENFA).

A preliminary choice then had to be made as to the example we would use. The types of transgenesis being considered by researchers (though whether for the short, medium, long or very long term is open to question, in view of the difficulties involved) aim to improve the various types of zootechnic performance, including modifications in milk composition (higher protein yields, higher cheese-making quality), genetic vaccination, muscle development, reduction of fatty deposits, enhanced use of food rations, improved wool growth, acquisition of independence from essential amino-acids in monogastric animals, changes in the composition of body fluids, improved prolificity, etc. Once useful genes have been identified and sequenced, and when techniques have become more effective, useful zootechnic characteristics often depend on QTL (Quantitative Trait Loci). The genome sequences involved are often located on several chromosomes and have a

cumulative effect on performance. Few transgenic animals have been born so far but experiments are under way.

We chose to draw on Canadian studies on the production of giant transgenic salmon expressing a foreign gene for the growth hormone, dubbed 'Sumotori salmon' after the Japanese wrestlers (Devlin 1994). In 1994, Canadian geneticists from the West Vancouver Department of Fisheries and Oceans (British Columbia), with two researchers from the US and Singapore, produced transgenic salmon which, within a year, weighed 11 times more than their counterparts of the same age. One individual's growth rate was 37 times higher than the norm. Experiments on fish had given disappointing results until this time, since they were based on transferring a mammalian gene. For these successful experiments, researchers used genetic material from salmon.

The growth hormone transgene was injected into 3000 fertilized eggs, whose development had been inhibited just after fertilization by micro-injection into the blastodisc. After one year, the transgene was effective on 6.2% of the surviving alevins (young salmon) which had an impressive growth rate. This genetic modification also accelerated the sexual maturation of the fishes that were capable of breeding and transmitting their growth capabilities to their offspring. The technique however is not yet fully controlled - some embryos do not survive the modification since as the transgene is placed randomly in the recipient genome it can upset the expression of other parts of the genome or the expression of the transgene itself.

The value of producing transgenic domestic animals to improve zootechnic performance is debatable in view of the effectiveness and long pedigree of conventional selection methods, but for animals that are closer to their natural state, like Salmonids, the benefits could be more spectacular. Although we wanted to develop a fictional situation, because of the uncertain state of the knowledge involved, it still had to be realistic so as not to reinforce students' perceptions of biotechnology as somehow magical and all-powerful.

In preparing the didactic approach, we also had to consider the conceptions of those at the receiving end (Simonneaux 1997b, Simonneaux and Bourdon 1998, L. Simonneaux 1999). In line with EIBE objectives, the target audience was made up of students in scientific and non-scientific secondary school education. The preparatory work turned out to rely on decision-making: choices had to be made, especially on the references to be introduced and on linguistic options (as noted by Chevallard, didactic preparation results in the textualization of knowledge).

We explored three different avenues to find and select the contents to be presented. First, the bibliographic references in the source publication printed in *Nature* in 1994 led us to the scientific articles published on this topic. Through these we were able to interview the French research teams working in this area at INRA to help us select the information to be presented on the Canadian experiment. After this, as we did not want to confine the process of didactic transposition to scientific knowledge alone, we decided to include professional references. This led us to interview the head of the French union of aquaculture breeders to find out more about the salmon industry, especially the comparative importance of farmed and fished salmon, the role of fish processors and co-operatives, the state of the market and its fluctuations, as well as the opinions of the union and the fish farmers on the production of transgenic fish, European and international policies in this respect, etc. This interview provided us

Table 2. Selected content.

Choice of scientific content	History of animal domestication and selection History of the transfer of the growth hormone gene Natural history of wild salmon The Canadian experiment: methods, results and problems
Possible repercussions mentioned in economic, ecological and legal fields etc.	Increased fish farm productivity Increasing over-production Technology transfers to the developing world and famine reduction Ecosystem disruption Loss of biodiversity Risks to human health Labelling and consumer reactions Patenting life forms Company monopolies, etc.

with the basis for a realistic, contextualized document (two Sumotori salmon farms have in fact just begun to operate in Europe).

In exploring each of these avenues, we listed the economic, ecological, ethical, legal and political repercussions as seen by those concerned. Drawing on this, we decided to introduce, as an illustration, an ecological risk that was mentioned in several articles and by some of the people we interviewed. The cages in which the 'domesticated' fish are kept are not always escape-proof, and it is believed that Norwegian fjords are full of the descendants of 'domesticated' fish that escaped from offshore cages during severe storms. Between 5% and 30% of the fish taken at sea are believed to have originated in fish farms. If these Sumotori salmon, which are capable of breeding and passing on their capacity for growth to their descendants, were to escape, they would seriously disrupt their ecosystems. The selected contents are presented in table 2. We then decided to use a role-play to 'stage' the contents of the document.

This role-play was an exercise in decision-making. The students were faced with a situation that was fictitious but could be readily imagined: they had to decide whether or not they agreed with the installation of a Sumotori salmon farm in a seaside village. They acted out the roles of people taking part in a public debate organized by the Mayor.

The document contents:

- background information about the history of domestication;
- the history of growth hormone gene transfer;
- description of the research done on the Sumotori;
- setting the scene;

> In a seaside village close to a fishing harbour, a fish farmer, Yann Le Goff, is planning to breed genetically modified salmon which grow more rapidly and become giant, called Sumotori (from the name of Japanese wrestlers). The local population is very concerned about this project. A group including fishermen, consumers, conservationists, and traditional fish farmers formed a committee to fight against this project. However, Yann le Goff rallied support from the owner of the canning factory and a part of the local council. The Mayor decided to organize a public debate with specialists in the field.

- cards describing the characters' and the observers' tasks;

  The characters are: the fish farmer who wants to open the transgenic fish farm, the owner of the canning factory who is interested in the project, a traditional fish farmer, a master fisherman, a 'with it' media studies student, a gastronome, a fishmonger, a leader of an environmental association, a researcher in fish physiology, a member of the Surf Rider Foundation which is concerned on ecological issues on the seaside, a young mother who is pro organic agriculture, the Mayor, an African PhD student in biotechnology.

- a pre-post-test to assess the evolution of the opinions of students concerning animal transgenesis research.

The object is to determine how far a given context might shape students' statements of opinion, by checking whether they were influenced by the different applications (medical, industrial, veterinary, etc.) and candidate species (domestic mammals, mice, fish).

From the linguistic point of view, choices had to be made on the lexical content and the textual register. We tried to adapt the scientific lexicon to our target students, or in other words, to reformulate the most abstruse lexical items. We also asked an economics expert to check both the quality of the economic arguments we had collected and the relevance of the economic terms used (added value, production costs, productivity, etc.). We drew on several different textual registers: the historical, scientific and contextual references supplied before the start of the role play are in an explanatory and descriptive register, while the description of the roles shifts from a descriptive to an argumentative register.

The preparatory didactic work involved a succession of adjustments. The didactic transposition process proved to be a process of didactic creativity, in which we were creating a unique teaching object by merging forms of scientific knowledge that are unstable and open to debate, with different aspects of a socio-professional situation. We made a great many adjustments in the course of selecting new information (Simonneaux *et al*. 1997).

Classroom protocol

Our study was conducted in two classes at the Agricultural Lycée in Auch (Gers *département* in southwest France), with students in their 2nd year of upper secondary vocational education geared to scientific subjects. All the dialogues were recorded on audio- and video-taped and transcribed in full.

The role-play situation

Before the role-play, a questionnaire assessed students' opinions about different research concerning animal transgenesis. After transgenesis was introduced with a presentation of the history of domestication and the background of growth hormone gene transfer, the role-play was presented. After the subject and interest of the role-play was presented, students expressed and justified their opinions on the setting up of a Sumotori farm.

Once the roles had been handed out, the students made up a list of questions that they would like to ask and expressed their arguments from their role's viewpoint. The students knew which individual characters were going to be involved in

Table 3. Experiment protocol.

Situation ❶ Role-play	*Situation ❷ Debate*
Pre-test opinions	ditto
Introduction to the history of domestication and chronology of the transfer of the growth hormone gene	ditto
Presentation of the role-play	Presentation of the debate
Written essay by students on the installation of a Sumotori fish farm (opinions to be supported)	ditto
Role-play performed	Debate
Post-test opinions	ditto
Written essay by students on the installation of a Sumotori fish farm (opinions to be supported)	ditto
Students explain under what circumstance(s) they might change their opinions	ditto

the role-play. The teacher distributed labels on which were written the names and the job/role specification of the participants. The observers organized themselves in order to gather together their observations.

When the role-play was underway, the teacher played the role of the Mayor - introducing the role-play and being responsible for the timing. At the end the teacher (the Mayor) asks the students to vote. At the end of the role-play, each individual participant expressed his/her opinion on the proposal and specified under what circumstances he/she would change their mind. After the role-play, students filled in the questionnaire on opinions again. The method and individual feelings were discussed. The decision-making process was analysed with the assistance of the observers.

The conventional debate situation

Apart from being based on a debate, this situation was organized in exactly the same way as the role-play. The experiment protocol is developed in table 3.

The teacher explains the debating situation by making an analogy with the citizens' conference that took place in France on the subject of biotechnology applications and sets out the aims of the debate, i.e., to argue out a decision, explaining that the first task is to discuss the economic, political, ecological and human health aspects of the issue. The teacher remains neutral, leaving the students to take up the various points spontaneously. If one or more aspects are forgotten, the teacher asks for their opinion on these. Next, the teacher prompts a debate on the various topics addressed in the role-play, unless they have already been taken up spontaneously by the students: increase in livestock productivity, increasing overproduction, technology transfers to the developing world, famine reduction, disruption of ecosystem balance, risks to human health, labelling and consumer reactions, patenting living organisms, company monopolies, and so on. As the debate proceeds, the teacher supplies the information that was contained in the role-play.

Method

The quantitative data from the study (pre- and post-test) were processed with Sphinx software to make calculations on the respondents' citations.

This paper will not go into the details of the analysis of the pre- and post-tests on students' opinions concerning research on different types of animal transgenesis. The results we observed were similar to those in a previous study conducted with the same method (Simonneaux 1999). Students' opinions varied according to the context and the research applications being considered. Medical applications were the most broadly accepted, followed by veterinary applications. Industrial applications (i.e. in food processing) were rejected. Opinions did not change between the pre-test and post-test stages. Rather, the paper is an analysis of students' statements before and after the role-play and debate.

Results

Opinions before and after the role-play

When students expressed a train of thought with arguments for and against, their final verdict, once it had been explicitly stated in conclusion to their reasoning, was taken into account.

Before the role-play, 11 students out of 18 stated that they were against the installation of a Sumotori salmon farm. Fifteen were against research aiming to produce these fish. This suggests that four students who had discussed the benefits and disadvantages of the fish farm in their responses without coming to an explicit conclusion were actually against the project. The same phenomenon was apparent after the role-play.

This requires some comment. In opinion surveys generally, as in the Swiss referendum for example, there is evidence that people tend to reject biotechnology applications but not biotechnology research. There is less opposition to research as such, which seems to be considered as reflecting a quintessentially human drive to push back the frontiers of ignorance. Therefore, if the research is rejected, the resulting applications are necessarily rejected as well. In addition, students question the status of biotechnology research, which they see as different from funda-

Table 4. Students' opinions before and after the role-play.

Students' opinions on Sumotori research before the role play	*Students' opinions on the installation of a Sumotori salmon farm before the role play*	*Students' opinions on Sumotori research after the role play*	*Students' opinions on the installation of a Sumotori salmon farm after the role play*
15 against 1 no reply	11 against 5 discussed arguments for and against 2 in favour under certain conditions 1 no reply	13 against 3 no replies	11 against 1 discussed arguments for and against 4 in favour under certain conditions 3 no replies
$n = 18$	$n = 18$	$n = 16$	$n = 16$

mental research where the aim is to produce new knowledge that is not necessarily applicable. Biotechnology research is seen as research with a purpose that will necessarily be applied eventually. Students' opinions before and after the role-play are presented in table 4.

Three students changed their minds on the installation of a Sumotori salmon farm. Their roles were as follows: one observer decided against the farm after the role-play, after having discussed its benefits and disadvantages; the student playing the role of Yann Le Goff, the fish farmer intending to set up a Sumotori farm, had been against the project before the role-play and subsequently came out in favour, subject to certain conditions (was the change of mind linked with the fact of acting out the part?); finally, the student acting the part of the 'foodie' opposing the farm stopped weighing the arguments for and against after the role-play, and came out against the project.

The students who discussed the arguments for and against thus confirmed the limits of conventional logic. Their verdicts were suspended between acceptance and refusal (see Giordan 1999, on this subject).

Opinions before and after the debate

Students' opinions before and after the debate are presented in table 5.

Four students changed their minds on the installation of a Sumotori salmon farm. It is important that the decision-making process includes listening to the views of antagonists (Kolstoe 2000).

From this point of view, the case of Marianne is interesting. Marianne's extreme positions had the effect of catalysing her colleagues' arguments. She was fundamentally in favour of the Sumotori salmon farm, and vigorously argued her case:

> It doesn't matter that much if a species disappears.
>
> In any case, producing countries will always be the richest ones; and we won't be giving to poor countries anyway.
>
> In any case we'll be discovering medicines to destroy what we've destroyed.

Table 5. Students' opinions before and after the debate.

Students' opinions on Sumotori research before the debate	*Students' opinions on the installation of a Sumotori salmon farm before the debate*	*Students' opinions on Sumotori research after the debate*	*Students' opinions on the installation of a Sumotori salmon farm after the debate*
14 against	10 against 1 in favour 2 discussed arguments for and against 2 in favour under certain conditions 2 against unless restrictions are imposed	13 against	14 against 1 in favour 1 in favour under certain conditions 1 against unless restrictions are imposed
$n = 17$	$n = 17$	$n = 17$	$n = 17$

> Bankrupt fish farmers will just have to do something else, you don't make an omelette without breaking eggs.
>
> We're not going to go back to the horse and cart are we! It's just too bad if there's unemployment.
>
> Cloning people might be useful: it might interest the army, or for work.

In every study that has been conducted so far, there has always been a small minority expressing ideas like these. It had been planned that the discussion leader (the teacher) would provide counter-arguments at the end of the debate to stimulate the discussion, if none had been forthcoming, but this turned out to be unnecessary.

Fields of reference used in arguments before and after the role-play

Students' arguments are based on what linguists refer to as *guarantors*. These may be associated with different disciplinary fields. The students' arguments were selected from the entire corpus and classified by field of reference, using an iterative approach. Some of the arguments could not be classified into a particular disciplinary field, a point to be returned to later. Fields of reference used before and after the role-play are presented in table 6.

The disciplinary fields called upon in support of arguments were economics, ecology, genetics, medicine and ethics. Political, legal and professional fields of reference did not appear.

Table 6. Number of arguments in the different fields of reference used before and after the role-play.

Arguments	*Before the role-play*	*After the role-play*
In favour	Positive effects on human health (1) Positive ethical effects (2) Positive economic effects (4)	Positive economic effects (2) Positive ethical effects (2)
Against	Negative effects on human health (6) Negative economic effects (6) Negative ecological effects (8) Negative genetic effects (1)	Negative ecological effects (8) Negative effects on human health (4) Negative economic effects (4)

Table 7. Number of arguments in the different fields of reference used before and after the debate.

Arguments	*Before the debate*	*After the debate*
In favour	Positive effects on science (2) Positive ecological effects (2) Positive economic effects (2)	Positive economic effects (1)
Against	Negative effects on human health (3) Negative economic effects (13) Negative ecological effects (5)	Negative ecological effects (5) Negative effects on human health (5) Negative economic effects (11) Negative effects on science (1) Negative political effects (3)

Table 8. Comparison between 'orders of importance' brought out in the role-play and the conventional debate.

	Role-play	*Debate*
Orders of importance in pre-test stage	Counter-arguments from the world of trade. Arguments from the worlds of civics, inspiration, trade and industry.	Counter-arguments from the world of trade Arguments from the worlds of civics, inspiration
Orders of importance in post-test stage	Counter-arguments from the world of trade. Arguments from the worlds of civics, inspiration, trade.	Counter-arguments from the world of trade Arguments from the worlds of civics, inspiration, trade

Fields of reference used in arguments before and after the debate

Fields of reference used before and after the debate are presented in table 7.

The disciplinary fields called upon in support of arguments were science, economics, ecology, politics and medicine. Legal, ethical, genetic and professional fields of reference did not appear.

Origins of justifications according to the 'order of importance' attached to the installation of a Sumotori salmon farm

Human actions (verdicts on objects of debate are considered here as actions) are structured around principles of justification. According to sociology of justification, what is important in justification processes is the 'order of importance' attached to objects of debate. I have selected, from the discourses, items showing the 'order of importance' attached to the installation of a Sumotori salmon farm, which indicate the 'world' of reference of students. Specific topics (Nature, quality, patrimony, 'disturbance' to the population) which could not be considered through the identification of arguments have been then analysed.

With reference to 'orders of importance', as seen in table 8, four 'worlds': trade, civics, industry and inspiration were identified. Supporting arguments from the 'world of civics' were in the majority. Here, the collective interest is predominant, through respect for legal forms. Supporting arguments from the 'world of civics' partly overlapped with medical concerns. There were connections with other topics, such as poverty, animal welfare, loss of the village heritage or social development in the village, and also with 'disturbance' to the population, 'the population might be disturbed by the fish farm, and the region could be upset by the person', and unemployment, 'it would be better to find jobs for hundreds more people; what about the small fish farmers?', the crucial problem of maintaining quality, consumer information and the need to establish tests and inspections.

The 'world of trade' was mostly called on in terms of opposition. Here, competition between the protagonists, through market forces, promotes the optimal use of resources. Profits are legitimate. This is where the students found most of their counter arguments. Commercial arguments that might have countered their own were rejected outright, to cut the discussion short, in line with the rules of rhetoric.

Of course it'll bring in a lot of money, but it'll bankrupt the small fish shops and the fishermen.

Debts from the high costs of installing the fish farm might ruin the fish farmer; and prices might crash because of large-scale production.

The 'world of inspiration' holds all that is sacred. Here, it was called on in references to nature. Establishing the Sumotori fish farm goes against nature. Man sees himself as a demigod. The techniques invented by people to serve their needs and culture will permanently disrupt the balance of nature. Nature must be left alone, as long as people do not intervene, it keeps in balance. Nature is the evolution of species that have adapted at random to a given environment. But the world is now dominated by humans and human culture, and species adaptation is under human control. We are trying to control the living world by artificial means, to assert ourselves as masters of the world, controlling all life; we want to become gods, we're tempted by eugenics, we want to adapt nature to our cultural, social and economic needs. Here, we are in a mythical register, as noted by Davallon (1991) in a study on representations of chemistry, a register that has nothing to do with scientific logic, but a lot to do with what might be called 'holy terror'. Controlling the living world means trying to control what is random, using technology to overcome chaos and our own fear of life and death.

Supporting arguments from the 'world of industry' were very much in the minority. Here, what is recognized as important is based on performance and method. Investment is linked with progress.

Ecological concerns are expressed in scientific terms (biodiversity, ecosystem, etc.) or in more commonly used terms (nature). Should these be classified in the 'world of civics', or the 'world of inspiration'? Or elsewhere? Arguments expressed in terms of biodiversity and ecosystems were placed in the 'world of civics', and those associated with nature in the world of inspiration. But perhaps we should look into the possible emergence of another 'world', in addition to those defined by Boltanski and Thévenot, a 'world' in which new social demands are being expressed in connection with the environment, a 'world of ecology'?

Analysis of argument sequences

The pattern of an argument rests on linkages that are made between the 'data' and a 'conclusion'. These linkages may be implicitly or explicitly supported (by the 'guarantor' and the' material') or opposed (by 'refutation' or by an 'exception').

The 'guarantor' fields of reference were described above. Some arguments include possible restrictions: some students are 'in favour if', 'in favour but', 'against unless', 'in favour as long as'.

Restrictions are brought in through various argument connectors, such as 'but, on the other hand, if it's true that, although, as long as', etc.

If it's true that it's useful to have reserves of animal protein.

As long as the fish has the same taste and flavour.

If there's only one breed involved.

As long as things don't get out of hand

Still, the aim is mostly economic.

Still, you can't say there's no risk.

Table 9. Circumstances associated with changes of mind after the role-play.

Student No	*Circumstances that might cause a change of mind*
1	If the fish farm had some use, for example feeding developing countries.
2	Only if the products are known to be 100% safe.
3	If there are none of the harmful effects mentioned above, in the long term.
4	I would not change my mind.
5	As long as the fish are watched, looked after.
6	Only if it's safe for nature, animals and consumers.
7	When they're perfect.
8	With very strict standards so there are no escapes, strict health services, continuous testing of the fishes' flesh...
9	If the scientific results are credible.
10	If it doesn't hurt competition from small businesses, if eating them is not harmful to health, when we see different positive results from the laboratories.
11	If the farm was small-scale.
12	Under no circumstances.
13	No reply.
14	Under no circumstances.
15	If there are problems that might harm fauna and flora, in the sea or on land.
16	No reply.
17	Under no circumstances.
18	Under no circumstances.

It is important to point out that 'guarantors' are not always explicit, and that conceptual misunderstandings (which arise when an item of information has not been properly understood) sometimes have an adverse effect on the quality of an argument. For six students, the 'uncertainty as to consequences' argument was the basis for recommending observance of the precautionary principle.

How students changed their minds

When asked after the role-play under what circumstances they would change their minds, five students stated that they would not change their minds under any circumstances: their opinions were definite. This question was asked to launch the students' discursive processes and to encourage them to justify their decisions.

As seen in table 9, these circumstances have to do with ecology, economics and science, and also relate to food problems in the developing world, food safety issues and human health. Safety and inspections are also mentioned.

As seen in table 10, these circumstances after the debate have to do with ecology and economics and also relate to food problems in developing countries, food safety issues and human health. Safety, control and inspections were also mentioned. Two of the students would feel happier about the project if the fish were sterile. Three students would not change their minds under any circumstances.

Table 10. Circumstances associated with changes of mind after the debate.

Student No	*Circumstances that might cause a change of mind*
1	If the fish farm raises no ecological risks and if it's a form of progress that doesn't have any economic consequences.
2	If the researchers and industrialists know exactly what they are doing.
3	If I had proof that there is no difference in quality between 'normal' salmon 'sumos', and if we could share out the surplus products to poor countries.
4	If I was given proof that there would be no ecological and health problems, maybe. But as long as there's no proof that these fish are healthy and of high quality, then I wouldn't mind.
5	I wouldn't change my mind.
6	If there were proof that there would be no effects on human health.
7	If we were sure it wouldn't cause any diseases and that it would help developing countries.
8	I'm totally in favour of continuing research, because the intentions are always good, but it's the next stage that's the most difficult. That's why if all the consequences were measured then maybe I'd think the Sumotori fish farms were a good thing. Consumers have to have information on where they come from and on their quality.
9	If the fish were sterile and couldn't get out of control.
10	If I were sure there were no risks and the fish are sterile.
11	Under no circumstances.
12	If research proves that there would be no consequences for humans, animals and nature (which we don't always think about).
13	No reply.
14	If there was no danger for humans and if people are fully in control of all spatial parameters. Which isn't possible.
15	If the farm had real advantages, from economic and ecological points of view (preserving wild species).
16	Under no circumstances.
17	Just possibly, if there were serious consequences for human organisms, for instance if there was a real threat to the human race.

Comparison of difficulties, advantages and disadvantages

These comparisons were made on the basis of teachers' comments, students' comments after the debate and from direct observation. The main obstacle concerning the role-play was the teachers' lack of familiarity with role-play practice, which was in fact one of the reasons for conducting this study. EIBE strongly recommends using role-play and teachers in northern Europe readily do so. Teachers need to be reassured and offered training sessions on organising role-plays, which are particularly well suited to interdisciplinary approaches.

The role-play assessment shows that the teacher was unable to remain neutral throughout in performing the Mayor's role. The following exchange shows that the students, when it came to voting, were aware of the impact of the teacher/Mayor on their classmates and were not taken in:

> Foodie: No, no, no, there's a fiddle going on. If the Mayor's being paid as well ...
>
> Mayor: My concern is for the development of our village.
>
> Foodie: Someone's getting a bribe, we'll take another vote. Right, who's in favour? Nobody. Who's against?

'Switched-on' communications student: If you'd done communications studies, Sir, you'd realize there's a boss here who keeps the meeting in order.

Foodie: So who's the boss?

Mayor: The Mayor of your village, Sir. I'm here because I'm responsible for our citizens' welfare. We'll take a final vote. Who's in favour of the project? Five. Who's neutral? Two.

Traditional fish farmer: No, I didn't raise my hand.

Fish physiology researcher: You're not allowed to influence people. Careful!

Mayor: Who's against? Seven.

The student playing the part of the communications student put her finger on the institutional relationship with the teacher.

Some students found it difficult to appropriate the information contained in the description of their roles. As observed by Kolstoe (2000), role-playing increases possibility to understand other people's point of view when you have to place yourself in their situation. But whether or not their interpretations were correct, the students kept strictly to the information supplied. Should they be asked to carry out some documentary research beforehand? The students were very enthusiastic about this activity, and said they did not feel ill at ease with it. However, some emphasised the difficulty of acting the part of a person they did not agree with. The observers felt frustrated at not being able to take part.

The teacher was keen to try the experiment again, and also said how interesting it had been to 'see the students in a different light that brings out their personalities'. The major problem concerning the debate here was the reserve of some students. Although they were asked several times for their opinions, seven students did not take part in the debate. When they were questioned afterwards, they stated that they were very interested but did not want to repeat opinions that had already been expressed, and in fact, in their written answers five of these students were the most loquacious on the subject. The quality of their argumentation, as measured by the number of arguments given, had improved between the pre and post debate stages. All the students expressed enthusiasm for the exercise.

The role of the teacher

As noted above, the role of the teacher is difficult. It is difficult to keep a neutral attitude during the debate and to promote an attitude of respect towards antagonists' opinions. And he/she is supposed to play an active role at the end of the debate by asking reflective questions to increase the students' awareness of the role and of the limits of scientific knowledge and of values issues brought into the debate. This aftermath of a discussion, whatever the form it takes (debate, role-play) is essential to the process: this is the stage which, as the different steps in an argument are highlighted *a posteriori*, helps students to achieve greater distance in relation to the topic - or, in other words, to adopt a metacognitive and 'meta-emotional' approach.

In this kind of issue, not only biotechnological knowledge but also societal consequences are to be taken into account. This point, as well as the unfamiliarity of science teachers with debate situation should promote the participation of humanities teachers.

Conclusion

The fundamental value of the didactic strategies concerned here lies in their potential for opening up the school environment to a dimension that reaches beyond the acquisition of knowledge, by socialising and contextualizing such knowledge and discussing the issues that arise from it. The fundamental goal is, therefore, to help students become full and active members of society.

Jiménez Aleixandre *et al*. (1998) have shown that the way students argue on environmental issues is different from what has been observed in more conventional 'scientific' subjects of surveys, insofar as the students used several different arguments. In this field, in their view, there is no single reference argument resting on an expert's point of view. Moreover, these authors highlight the importance of values in students' arguments (e.g. pragmatism versus utopia, economics versus ecology). Their research draws on the method developed by Toulmin (1958) to analyse arguments. In our assessments, we also observed the variety in students' arguments. However, it was not so much the number of arguments that increased as the persuasiveness of their discourse. The same arguments were better supported and expressed more strongly after a debate. Our hypothesis is that in informed and committed decision making, individuals tend to use fewer arguments, feeling that a few decisive arguments are enough.

Classroom debates encourage student participation in discourse on science. We believe that students take a more active interest in controversy than in the past, perhaps because of generally increasing suspicion towards scientific experts. By using a different theoretical framework (based on the sociology of justification), we were able to analyse the values supporting students' arguments in terms of the 'orders of importance' they attached to them. This framework, which was borrowed from sociological and economics research, seems to us to be very promising for studies on justification and argument situations and for studies on classroom interaction in general. It enabled us to analyse responses that were not directly connected to disciplinary fields and, especially, to describe more fully what are often simply referred to as the 'values' underlying arguments and decision making.

There is a need to pursue this analysis of debating situations in the classroom, looking at arguments in detail as they develop, from the linguistic theory angle but also in the light of the economics of importance theory. It should be possible to identify, as discussions proceed, the quality and nature of students' knowledge in the disciplines which they call upon, and - using the economics of importance theory - the 'order of importance' underlying their discourse.

Numerous research questions emerge: how do social representations determine the principles of justification used by individuals? Can the nature of debates be enriched by defining the principles of justification used by each protagonist and making comparisons between those given by scientists, agro-chemicals companies and students? Does any possibility of a 'consensus' emerge between protagonists with seemingly opposing points of view? Or of a 'higher common principle' on which a compromise might be built? Do the social and identity characteristics of students determine the 'worlds' to which they belong?

Various research efforts in didactics involve evaluating the quality of students' argumentation. However, the criteria used to evaluate the quality of argumentation have by no means been firmly defined. What is sound argumentation? There is no obvious answer: is it based on several arguments, or on the discussion of contra-

dictory arguments, or on one decisive argument? Different analytical tables have been suggested by various researchers, and the subject is by no means exhausted.

Using the economics of importance theory as a framework has shifted the focus of research in the area of classroom argumentation. The idea here is not to supply criteria for evaluating its quality, but rather to analyse the 'order of importance' attached to the biotechnology applications being discussed, and how they affect the making of decisions on the crucial issues these applications raise for society.

This study draws on observations of real classroom situations rather than on experimental situations that are set up specifically for research purposes. We found few significant differences between the arguments put forward by the students in each variant (role-play and debate). However, among all the studies we have conducted so far, this was the first in which we observed changes of opinions. Our previous results had not been particularly surprising insofar as opinions - as the foundations of social representations - are not easily shifted. Before and after a number of formal and informal learning sequences (visits to exhibitions), we had always found knowledge being appropriated without any changes of opinion. But in these situations, the students had not been asked to discuss issues orally. Could it be that it is in expressing points of view and being confronted with opposing arguments that students clarify their thoughts on a given subject, as asserted by Barnes and Todd (1977) and Lewis *et al*. (1999)? If so, the didactic strategy involving class discussions, whether through role play or debate, would seem to be a useful way of helping students to develop their arguments. Although the expressions used by the students sometimes showed a lack of conceptual understanding of the biotechnology examples used, this did not prevent them from taking an active part in discussions and argument.

Science teaching has only made a modest contribution so far to the way opinions are formed (at least, in France). Teaching students to identify and assess opinions and to form their own well thought-out opinions on a complex problem of both scientific and social importance should logically be considered an essential aspect of scientific education and the acquisition of scientific literacy.

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